CHAPTER 3

Descriptive Analytics: Nature of Data, Big Data & Statistical Modeling

From raw facts to analytics-ready insight — and the statistics that turn data into decisions.

ISM 6404 · Business Intelligence & Analytics

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Adapted from Sharda, Delen & Turban, Analytics, Data Science, & Artificial Intelligence: Systems for Decision Support (Pearson), Ch. 3.

ROADMAP

What This Chapter Covers

LO 3.1 The nature of data in BI, data science & analytics

LO 3.2 A simple taxonomy to classify types of data

LO 3.3 Methods to process real-world data for analytics

LO 3.4 What Big Data is and how it changes analytics

LO 3.5 The principles & business drivers of Big Data

LO 3.8 Statistical modeling and its link to analytics

LO 3.9 Descriptive & inferential statistics in analytics

Enabling technologies (Hadoop, MapReduce, NoSQL) and batch/stream analytics — LO 3.6–3.7 — are covered in a separate session.

From Data to Wisdom

Data is a collection of facts

— numbers, words, images — obtained through experiences, observations, and experiments.

The lowest level of abstraction

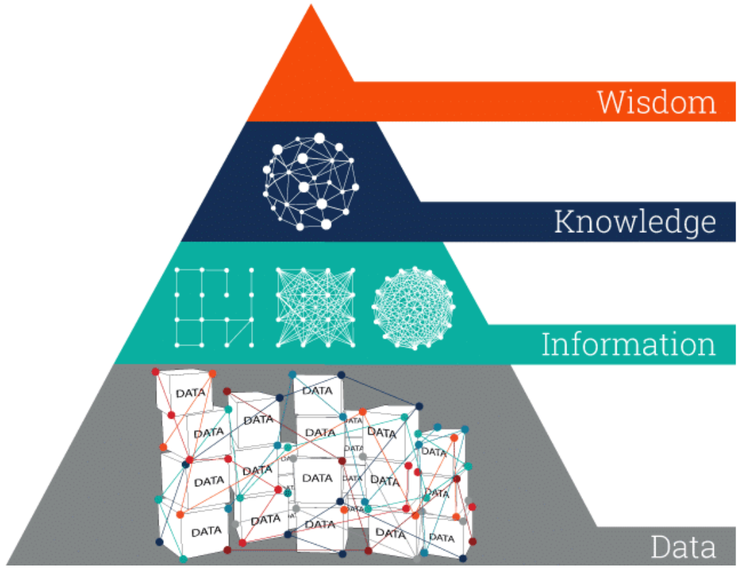
Data is the source from which information, knowledge, and ultimately wisdom are derived.

Each step up the pyramid adds value

Data → Information → Knowledge → Wisdom: every level answers a question the level below cannot.

Garbage in, garbage out (GIGO)

Data quality and data integrity are non-negotiable prerequisites for accurate analytics.



Each step up the pyramid answers questions about and adds value to the initial data.

Sharda, Delen & Turban (Pearson), §3.1 · Figure: Data-to-Knowledge pyramid

Is Your Data Analytics-Ready?

Ten metrics determine whether data can be trusted for decision making:

1. **Source reliability**
   Do we trust and believe this source?
2. **Content accuracy**
   Is it the right data for the job?
3. **Accessibility**
   Can we get it when we need it?
4. **Security & privacy**
   Protected from theft & misuse (HIPAA)?
5. **Richness**
   Enough dimensionality to model?
6. **Consistency**
   Same labels, units, and dimensions?
7. **Currency / timeliness**
   Is the data up to date?
8. **Granularity**
   Detailed enough for the outcome?
9. **Validity**
   The right dimensions for the result?
10. **Relevancy**
    Right variables — irrelevant = omitted

Classifying the Types of Data

Structured

Resides in fixed fields — databases & spreadsheets. Mined directly by analytics algorithms.

Categorical → Nominal & Ordinal

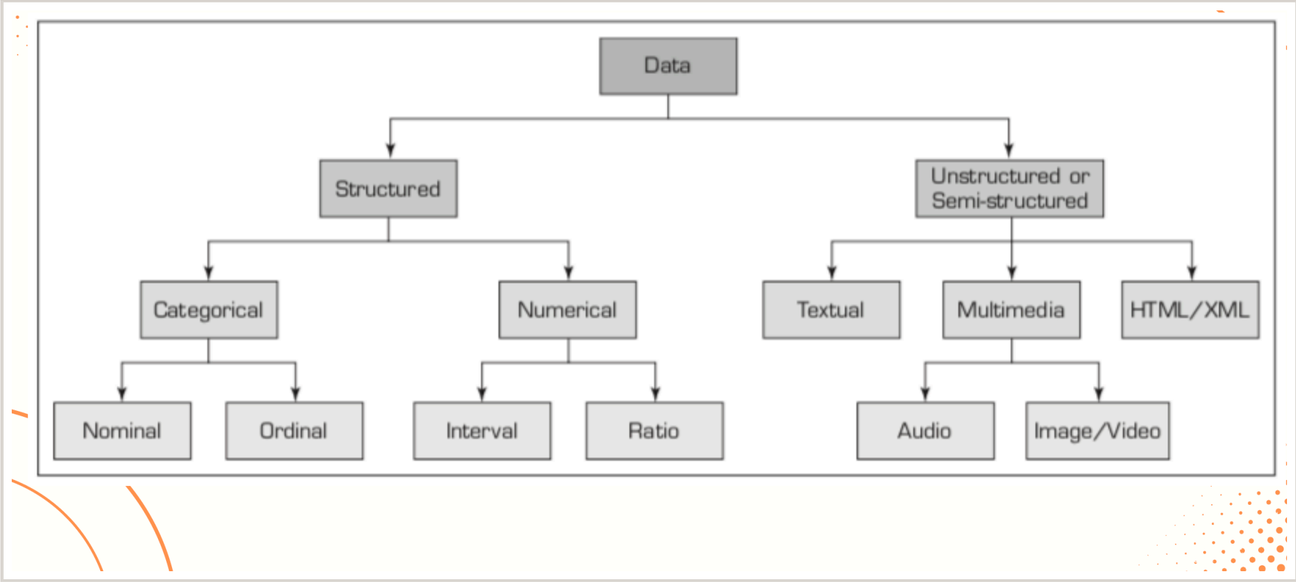
Labels (nominal) and rank-ordered codes (ordinal).

Numerical → Interval & Ratio

Measured values; ratio has a true zero.

Unstructured / Semi-structured

Text, multimedia, and web formats — needs text/Web mining first.



Sharda, Delen & Turban (Pearson), §3.2 · Figure 3.2 — Taxonomy of data

Structured vs. Unstructured Data

STRUCTURED

Nominal

Labels with no order — marital status, yes/no, true/false.

Ordinal

Ranked codes — credit score, age group, education level.

Interval

Measured on a scale, no true zero — temperature (°C).

Ratio

Numeric with true zero & equal distances — weight, length, time.

UNSTRUCTURED / SEMI-STRUCTURED

Textual

Customer reviews & documents → analyzed by text mining.

Multimedia

Video, audio, images (dogs vs. cats) → ML / prediction.

XML / JSON / HTML

Semi-structured formats for exchanging data in web apps.

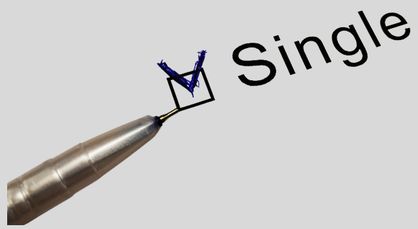
Roughly 80–85% of all organizational data is unstructured or semi-structured.

STRUCTURED DATA

CATEGORICAL

NOMINAL

LABELS:
MARITAL STATUS (MARRIED, SINGLE, DIVORCED)
TRUE/FALSE, YES/NO, WHITE/BLACK/LATINO



ORDINAL

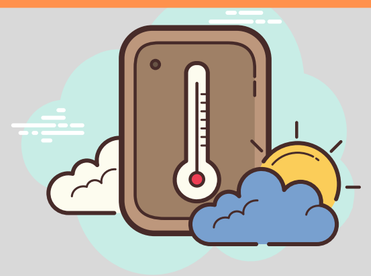
CODES ASSIGNED AS LABELS REPRESENTING RANK ORDER:
CREDIT SCORE, AGE GROUP, EDUCATIONAL LEVEL



NUMERICAL

INTERVAL

ABILITY TO MEASURE AT INTERVAL SCALE:
TEMPERATURE ON THE CELSIUS SCALE



RATIO

ABILITY TO MEASURE ON NUMERICAL SCALE, HAS EQUAL DISTANCES BETWEEN VALUES AND TRUE ZERO:
WEIGHT IN KILOGRAMS, LENGTH, TIME



UNSTRUCTURED OR SEMISTRUCTURED DATA

Text mining / Web Mining
Machine Learning / Prediction

TEXTUAL

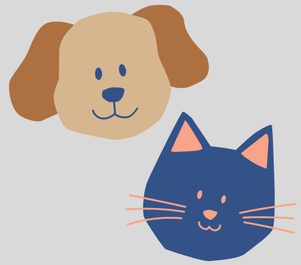
MEASURE REASONS FOR CUSTOMERS LOYALTY (SCAN ONLINE CUSTOMER REVIEWS)



MULTI MEDIA

YOUTUBE VIDEOS, PODCASTS (AUDIO), PICTURES (DOGS VS CATS)





XML/ JSON

TRANSMITTING DATA IN WEB APPLICATIONS (E.G., SENDING SOME DATA FROM THE SERVER TO THE CLIENT, DISPLAYS IT ON WEB PAGE, OR VICE VERSA).



The Art & Science of Preprocessing

1. Data Consolidation

Collect, select, and integrate the relevant data — including data blending across sources.

2. Data Cleaning

Impute missing values, reduce noise, and eliminate duplicates (a.k.a. data scrubbing).

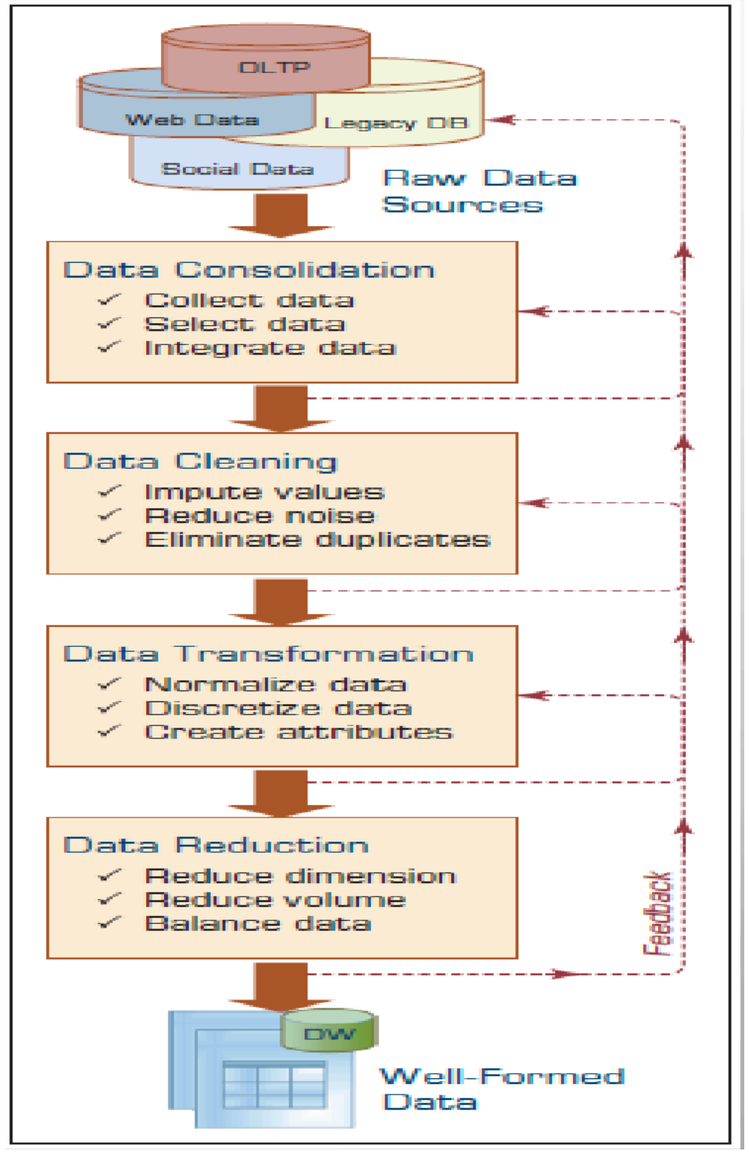
3. Data Transformation

Normalize, discretize, and create new attributes so variables are comparable.

4. Data Reduction

Reduce dimensions and volume; balance / stratify to produce well-formed data.

Often the least enjoyable yet longest phase of an analytics project — and the most decisive for results.



Sharda, Delen & Turban (Pearson), §3.3 · Figure 3.3 — Data preprocessing steps

Big Data & What Makes It Work

The “V’s” that define Big Data

Volume scale of data	**Variety** many formats	**Velocity** speed of flow
Veracity trust (IBM)	**Variability** shifting meaning	**Value** the payoff

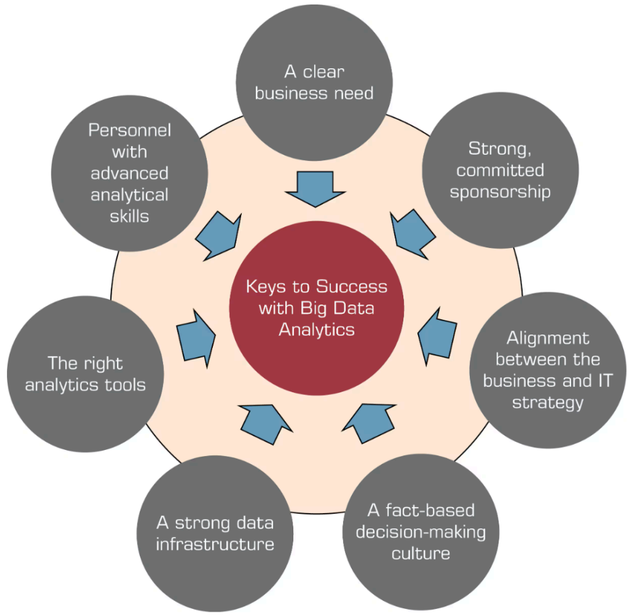
Big Data exceeds the capacity of traditional tools to capture, store, manage, and analyze it within a tolerable time — scaling from kilobytes to petabytes and beyond.

Critical success factors (§3.5):

Clear business need · executive sponsorship · business–IT alignment · a fact-based decision culture · a strong data infrastructure.

Figure 3.8

Critical Success Factors for Big Data Analytics.



Watson, H. (2012). The requirements for being an analytics-based organization. Business Intelligence Journal, 17(2), 42-44

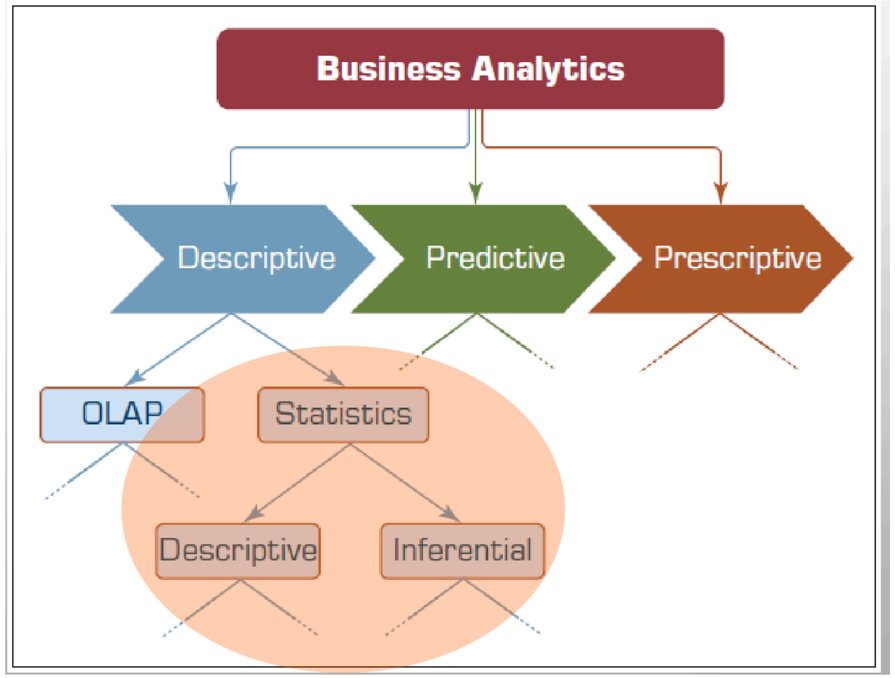
Statistics Behind Business Analytics

Statistics — a collection of mathematical techniques to characterize and interpret data.

Descriptive statistics describe the data as it is — the foundation of descriptive analytics.

Inferential statistics draw conclusions about a population from a sample — the basis of regression.

The OLAP / data-warehouse branch and statistics together power descriptive analytics.



Sharda, Delen & Turban (Pearson), §3.8 · Figure 3.11 — Statistics & descriptive analytics

Measures of Central Tendency

Mean

The arithmetic average of all values:

$$\overline{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Most common, but sensitive to outliers.

Median

The middle value when data is rank-ordered.

Robust — unaffected by extreme values.

Mode

The most frequently occurring value.

Useful for categorical data.

Rule of thumb: when data is skewed or contains outliers, the median describes the "typical" value better than the mean.

Measures of Dispersion

Measures of dispersion (it may also be called measures of spread or decentrality) are the mathematical methods used to estimate or describe the degree of variation in a given variable of interest. They are a representation of the numerical spread of a given data set. **To describe this dispersion, a number of statistical measures are developed;** the most notable ones are **range, variance, and standard deviation** (and also quartiles and absolute deviation).

One of t**he main reasons why the measures of dispersion/spread of data values are important** is the fact that it gives us a framework within which we can judge the central tendency—**gives us the indication of how well the mean (or other centrality measures) represents the sample data.**
Also, in research, it is often perceived as a positive sign to see a small variation within each data sample, as it may indicate **homogeneity, similarity, and robustness** within the collected data.

Range

The range is perhaps the simplest measure of dispersion. It is the difference between the largest and the smallest values in a given data set (i.e., variables) maximum minus minimum.

Variance

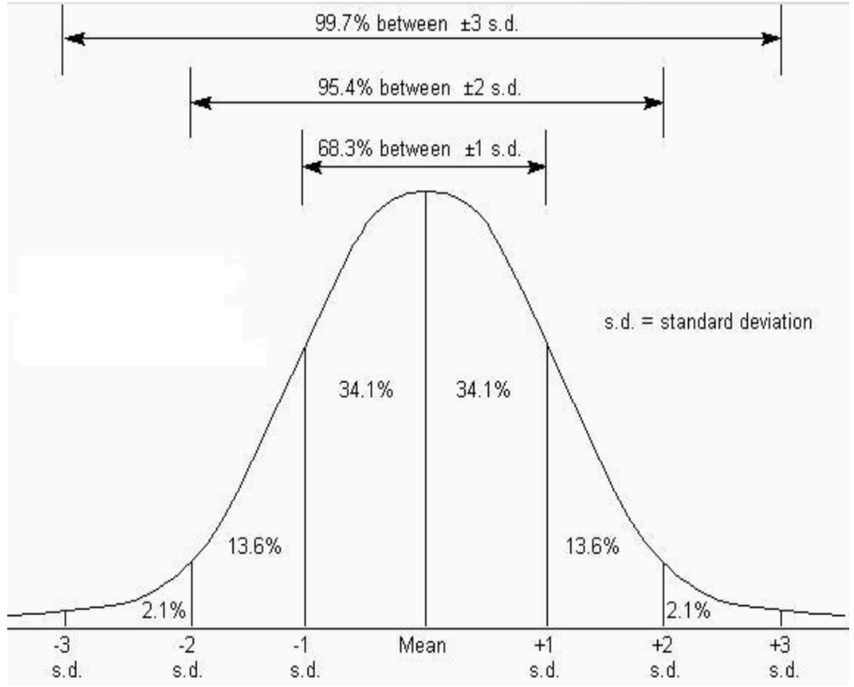
Measure how spread out the data is. Average squared deviation from the mean: $s^2 = \Sigma(x_i - \bar{x})^2 \,/\, (n - 1)$.

Standard deviation

The square root of variance — spread in the data's own units.

Mean absolute deviation

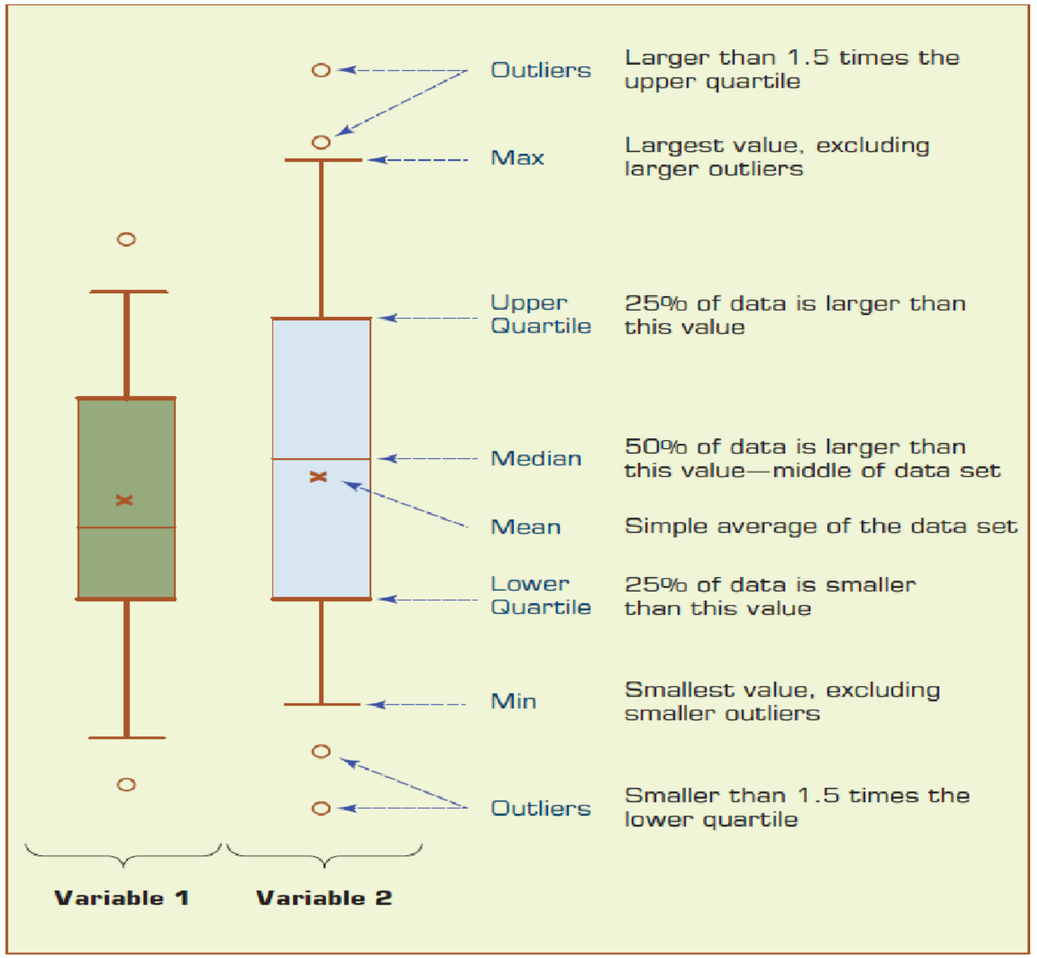
Average absolute distance from the mean (MAD).



Sharda, Delen & Turban (Pearson), §3.9 · Figure 3.12 — Box-and-whiskers plot

DESCRIPTIVE STATISTICS MEASURES OF DISPERSION

- QUARTILES
- • BOX-AND-WHISKERS PLOT
  - A.K.A. BOX-PLOT
  - VERSATILE / INFORMATIVE



The Shape of a Distribution

Skewness — a measure of asymmetry.

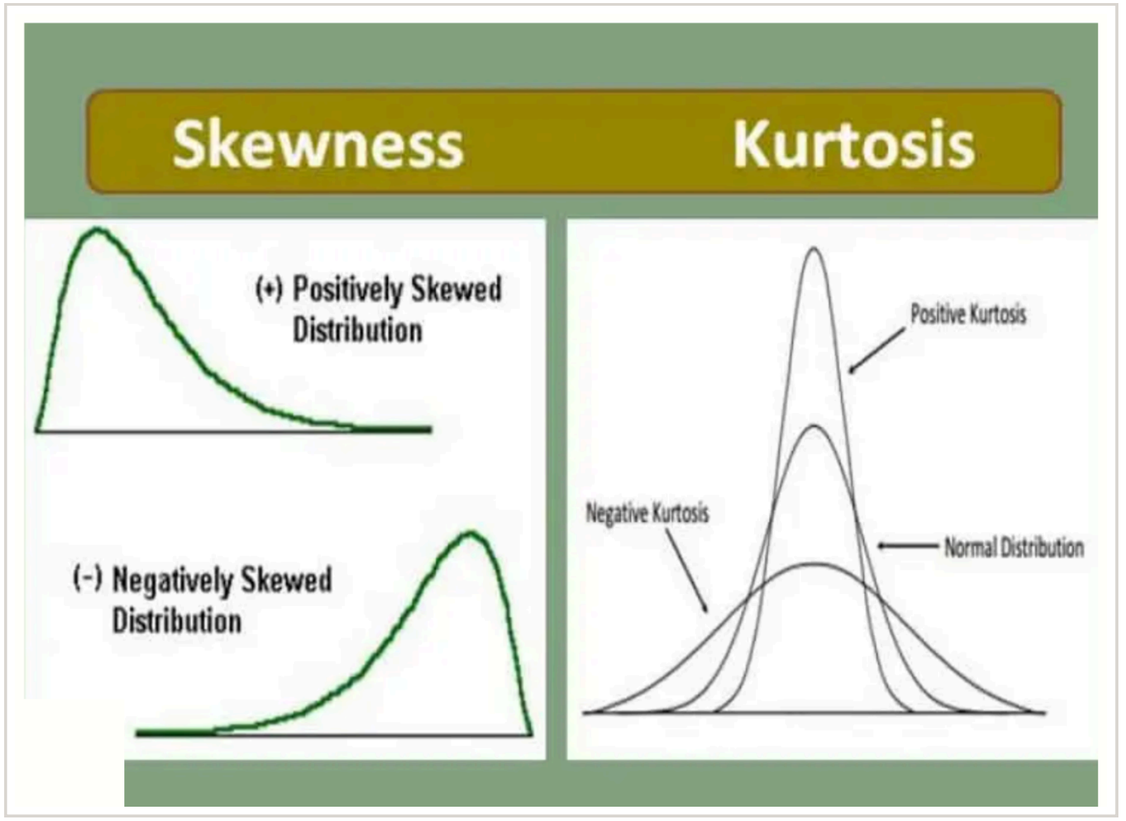
Positive (right) skew: long tail on the right; a few large values pull the mean above the median.

Negative (left) skew: long tail on the left; a few small values pull the mean below the median.

Kurtosis — the peakedness of the distribution.

Positive = tall and peaked; negative = flat and spread. A normal distribution has a kurtosis of 3.

A histogram is the quickest way to see both skewness and kurtosis at a glance.



Regression Modeling

Correlation vs. regression

Correlation measures the strength of a relationship; regression models how one variable affects another.

Simple vs. multiple

Simple: $y = \beta_0 + \beta_1 x$. Multiple: $y = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$.

Ordinary least squares (OLS)

Fits the line that minimizes the sum of squared errors.

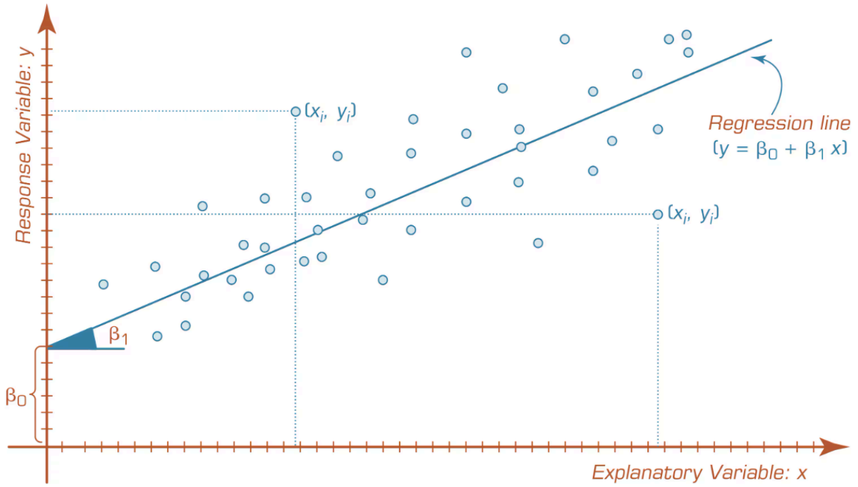
Is the model good enough?

R^2 close to 1, p-values close to 0, and low error (RMSE, MAE, MSE).

Process: data assessment → model fitting → model assessment → deployment.

Figure 3.17

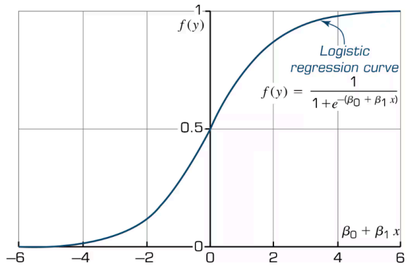
A Scatter Plot and a Linear Regression Line.



Logistic Regression

Figure 3.19

The Logistic Function.



The logistic function, $f(y)$ in Figure 3.19, is the core of logistic regression, which can only take values between 0 and 1. The following equation is a simple mathematical representation of this function:

$$f(y) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

A binary target

The output is a yes/no (binomial) class — not a numeric value as in linear regression.

A classification workhorse

Logistic regression is a very popular, statistically sound, probability-based classification algorithm that employs supervised learning. It was developed in the 1940s as a complement to linear regression and linear discriminant analysis methods. It has been used extensively in numerous disciplines, including the medical and social sciences fields.
It differs from linear regression with one major point: its output (response variable) is a class as opposed to a numerical variable. That is, whereas linear regression is used to estimate a continuous numerical variable, logistic regression is used to classify a categorical variable.

The logistic (S-curve)

Maps any input to a probability between 0 and 1, which is then mapped to a class.
In predictive analytics, logistic regression models are used to develop probabilistic models between one or more explanatory/predictor variables (which may be a mix of both continuous and categorical in nature) and a class/response variable (which may be binomial/binary or multinomial/multiclass).

Key linear-regression assumptions: linearity · independence of errors · normality · constant variance · no multicollinearity.

Sharda, Delen & Turban (Pearson), §3.9 · Figure 3.19 — The logistic function

LOGISTIC REGRESSION MODELING

- A VERY POPULAR STATISTICS-BASED CLASSIFICATION ALGORITHM
- EMPLOYS SUPERVISED LEARNING
- DEVELOPED IN 1940S
- THE DIFFERENCE BETWEEN LINEAR REGRESSION AND LOGISTIC REGRESSION
    - IN LOGISTIC REGRESSION OUTPUT/TARGET VARIABLE IS A BINOMIAL (BINARY CLASSIFICATION) VARIABLE (AS OPPOSED TO NUMERIC VARIABLE)

Classify Wine Types based on the Chemical Composition Properties

Type	Alcohol	Malic_Acid	Ash	Ash_Alcalinity	Magnesium	Total_Phenols	Flavanoids	nflavanoid_Pher	Proanthocyanins	Color_Intensity	Hue	OD280_OD315	Proline
A	13.73	4.36	2.26	22.5	88	1.28	0.47	0.52	1.15	6.62	0.78	1.75	520
A	13.45	3.7	2.6	23	111	1.7	0.92	0.43	1.46	10.68	0.85	1.56	695
A	12.82	3.37	2.3	19.5	88	1.48	0.66	0.4	0.97	10.26	0.72	1.75	685
A	12.52	2.43	2.17	21	88	2.55	2.27	0.26	1.22	2	0.9	2.78	325
B	11.76	2.68	2.92	20	103	1.75	2.03	0.6	1.05	3.8	1.23	2.5	607
B	11.41	0.74	2.5	21	88	2.48	2.01	0.42	1.44	3.08	1.1	2.31	434
B	12.08	1.39	2.5	22.5	84	2.56	2.29	0.43	1.04	2.9	0.93	3.19	385
B	11.03	1.51	2.2	21.5	85	2.46	2.17	0.52	2.01	1.9	1.71	2.87	407
C	11.82	1.47	1.99	20.8	86	1.98	1.6	0.3	1.53	1.95	0.95	3.33	495
C	13.28	1.64	2.84	15.5	110	2.6	2.68	0.34	1.36	4.6	1.09	2.78	880
?	13.05	1.65	2.55	18	98	2.45	2.43	0.29	1.44	4.25	1.12	2.51	1105
?	13.07	1.5	2.1	15.5	98	2.4	2.64	0.28	1.37	3.7	1.18	2.69	1020
?	14.22	3.99	2.51	13.2	128	3	3.04	0.2	2.08	5.1	0.89	3.53	760
?	13.56	1.71	2.31	16.2	117	3.15	3.29	0.34	2.34	6.13	0.95	3.38	795

REFERENCES & KEY TERMS

Sources for This Chapter

Primary source

Sharda, R., Delen, D., & Turban, E. Analytics, Data Science, & Artificial Intelligence: Systems for Decision Support. Pearson — Chapter 3, “Descriptive Analytics I: Nature of Data, Big Data, and Statistical Modeling.”

Supporting reference

Watson, H. (2012). The requirements for being an analytics-based organization. Business Intelligence Journal, 17(2). · HIPAA — Health Insurance Portability and Accountability Act.

Figures reproduced from the textbook (§3.1–3.9) for educational use in ISM 6404.

KEY TERMS

DIKW pyramid · GIGO · analytics-ready data · structured / unstructured data · nominal · ordinal · interval · ratio · data preprocessing · data blending · Big Data · the V’s · descriptive vs. inferential statistics · mean · median · mode · variance · standard deviation · skewness · kurtosis · correlation · OLS regression · R^2 · logistic regression